Listing Outcomes and Two-Stage Events

Name _____ Date _____ No calculator Version 1

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KEY WORDS outcome *(putanga)* · sample space *(mokowā tīpako)* · two-stage event

A Listing outcomes ●○○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

1. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3 is spun.

c) You choose a drink (juice J, water W, milk M) and a snack (apple A, biscuit B).

d) A coin is tossed three times (H, T).

2. How many outcomes are there? Multiply the number of choices at each stage.

a) 3 shirts and 4 pairs of shorts

b) Two dice are rolled

c) A coin is tossed and a dice is rolled

d) 4 mains, 3 desserts and 2 drinks

B Sample space tables ●●○

Remember: A table shows every outcome of two events once. Count the cells that fit, then divide by the total number of cells.

3. Two dice are rolled and the two numbers are added. Complete the table to show every possible total.

+	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

4. Use your table. Find the probability that the total is:

a) 7 _____ **b)** 10 _____ **c)** more than 9 _____

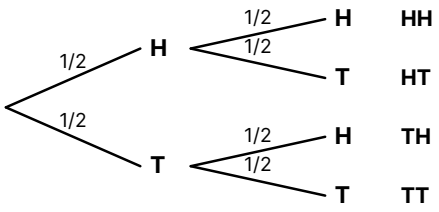
d) an even number _____ **e)** less than 4 _____ **f)** a multiple of 3 _____

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C Tree diagrams ●●●

> **Remember:** Multiply along the branches to find the probability of one path. Add the probabilities of the paths that fit.

5. A coin is tossed twice. The tree diagram shows the outcomes.

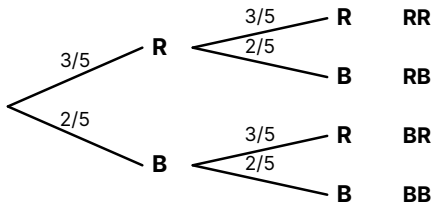


a) Find the probability of two heads.

b) Find the probability of one head and one tail, in either order.

c) Find the probability of at least one tail.

6. A bag holds 3 red and 2 blue counters. A counter is picked, its colour is noted and it is put back. Then a second counter is picked.



a) Find the probability that both counters are red.

b) Find the probability that both counters are blue.

c) Find the probability of one red and one blue, in either order.

d) Find the probability that the two counters are the same colour.

D Problems ●●●

7. A spinner numbered 1 to 4 is spun twice.

a) Find the probability that both spins give the same number.

b) Find the probability that the two numbers add to 5.

8. A family has two children. Each child is equally likely to be a boy or a girl. Find the probability that the family has one boy and one girl.

9. The probability of rain on Saturday is 0.3. The probability of rain on Sunday is 0.5. The two days do not affect each other.

a) Find the probability that it rains on both days.

b) Find the probability that it does not rain on either day.

★ Extension

10. A bag holds 3 red and 2 blue counters. Two counters are picked, one after the other. This time the first counter is not put back.

a) Find the probability that both counters are red.

b) Find the probability of one red and one blue, in either order.

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Listing Outcomes and Two-Stage Events

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS outcome *(putanga)* · sample space *(mokowā tīpako)* · two-stage event

A Listing outcomes ●○○

Remember: Be systematic: keep the first choice fixed and go through every second choice, then move on to the next first choice.

1. List all the possible outcomes. Use the letters and numbers given.

a) A coin is tossed twice (H, T).

b) A coin is tossed (H, T) and a spinner numbered 1, 2, 3, 4 is spun.

c) You choose a main (pie P, burger B, salad S) and a dessert (fruit F, cake C).

d) A coin is tossed three times (H, T).

2. How many outcomes are there? Multiply the number of choices at each stage.

a) 5 shirts and 3 pairs of shorts

b) Two dice are rolled

c) A coin is tossed and a dice is rolled

d) 3 mains, 5 desserts and 2 drinks

B Sample space tables ●●○

Remember: A table shows every outcome of two events once. Count the cells that fit, then divide by the total number of cells.

3. Two dice are rolled. The smaller number is subtracted from the larger number. Complete the table to show every possible difference.

–	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

4. Use your table. Find the probability that the difference is:

a) 0 _____ **b)** 1 _____ **c)** more than 3 _____

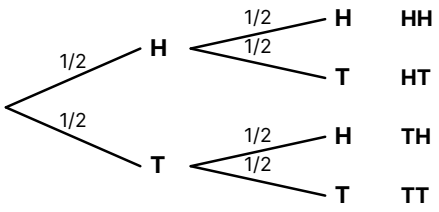
d) an even number _____ **e)** less than 2 _____ **f)** 3 or more _____

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C Tree diagrams ●●●

> **Remember:** Multiply along the branches to find the probability of one path. Add the probabilities of the paths that fit.

5. A coin is tossed twice. The tree diagram shows the outcomes.

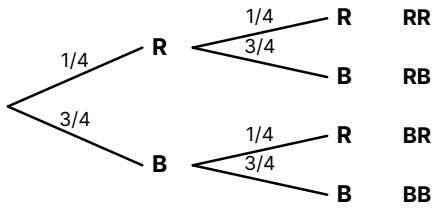


a) Find the probability of two heads.

b) Find the probability of one head and one tail, in either order.

c) Find the probability of at least one tail.

6. A bag holds 1 red and 3 blue counters. A counter is picked, its colour is noted and it is put back. Then a second counter is picked.



a) Find the probability that both counters are red.

b) Find the probability that both counters are blue.

c) Find the probability of one red and one blue, in either order.

d) Find the probability that the two counters are the same colour.

D Problems ●●●

7. A spinner numbered 1 to 5 is spun twice.

a) Find the probability that both spins give the same number.

b) Find the probability that the two numbers add to 6.

8. A family has two children. Each child is equally likely to be a boy or a girl. Find the probability that the family has one boy and one girl.

9. The probability of rain on Saturday is 0.4. The probability of rain on Sunday is 0.2. The two days do not affect each other.

a) Find the probability that it rains on both days.

b) Find the probability that it does not rain on either day.

★ Extension

10. A bag holds 4 red and 2 blue counters. Two counters are picked, one after the other. This time the first counter is not put back.

a) Find the probability that both counters are red.

b) Find the probability of one red and one blue, in either order.

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